

Amirhossein Masoudnezhad

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Profile

Master's student in Astrophysics and Cosmology with research experience in observational astronomy, cosmological simulations, and high-performance computing. Experienced in scientific programming, numerical methods, HPC workflows, and scientific software development. Interested in computational cosmology, large-scale simulations, and HPC-enabled research.

Education

M.Sc. in Astrophysics & Cosmology

2024–Present

University of Bologna

Relevant Coursework: Multi-Wavelength Astrophysics Lab., Practical Statistics, High-Performance Computing, Advanced Cosmology

B.Sc. in Physics

2019–2022

Sharif University of Technology

Relevant Coursework: Cosmology, Quantum Mechanics, Mathematical Physics, Machine Learning

B.Sc. in Electrical Engineering

2016–2019

Sharif University of Technology

Relevant Coursework: Signals and Systems, Communication Systems, Electronics

Research Interests

- Computational Cosmology and Large-Scale Structure
- Cosmological Simulations and Numerical Methods
- High-Performance Computing and Scientific Software Development
- Computational Astrophysics and Astronomical Data Analysis

Publications

- **Dusty stellar sources classification by implementing machine learning methods based on spectroscopic observations in the Magellanic Clouds.** The Astrophysical Journal, 2025
- **Spectral identification and classification of dusty stellar sources using spectroscopic and multiwavelength observations through machine learning.** IAU Symposium 368, 2023
- **Machine Learning Approach for the Spectral Classification of Dusty Stellar Sources in the Magellanic Clouds.** EAS Publications Series, 2024

Research Experience

Zoomed Initial Conditions (ZInCo)

Supervised by Prof. Marco Baldi, University of Bologna

Extended the ZInCo code for multi-resolution cosmological initial conditions, with improved particle merging and resolution management. Tested and validated the implementation across different simulation configurations.

ML Classification of Dusty Stellar Sources

Supervised by Prof. Atefeh Javadi, IPM

Developed machine-learning methods for classifying dusty stellar sources using spectroscopic and multi-wavelength observations, with automated data-analysis and classification pipelines.

Course Projects

Gadget-2 N-Body Simulations

High Performance Computing in Astrophysics & Cosmology

Ran cosmological N-body simulations on multi-core HPC clusters and evaluated strong-scaling performance and parallel efficiency.

Optical Spectroscopy of Globular Cluster Giant Stars

MWL – Optical Module

Reduced and analyzed high-resolution optical spectra to measure radial velocities and stellar abundances, validating results against reference cluster values using SLAB and SALVADOR.

Observation Planning for Bootes II RGB Stars

MWL – Optical Module

Designed an observing strategy for RGB stars in an ultra-faint dwarf galaxy, optimizing instrument selection, exposure times, and observing conditions.

Calibrating and Imaging VLA Data of RBS797

MWL – Radio Module

Calibrated and imaged VLA continuum data using CASA, measuring radio source properties and spectral characteristics.

Fermi-LAT Analysis of TXS 0506+056

MWL – High Energy & X-ray Module

Analyzed five years of Fermi-LAT gamma-ray data using `Fermipy`, generating light curves and spectral energy distributions to study source variability.

Scientific Software & HPC

ClusterPilot – Mobile HPC Management Platform

Designed and developed an iOS application to manage HPC clusters. Provides mobile access to SLURM job monitoring and submission, cluster file management, and SBATCH script configuration through a centralized API layer.

Technical Skills

Scientific Computing & HPC

Python (NumPy, SciPy, Astropy), C/C++, Fortran, Bash, MPI, Git, Linux/HPC environments

Astronomical Data Analysis

Optical Spectroscopy (SLAB, SALVADOR), Radio Astronomy (CASA), Gamma-ray Analysis (Fermipy, Fermi-LAT), observation planning, calibration, time-series analysis

Machine Learning

Supervised and unsupervised learning, neural networks, deep learning

Software Development

Swift/SwiftUI, Flutter/Dart, PHP, SQL, MongoDB, REST APIs, back-end development

Honors & Awards

- Ranked **103** among 160,000+ participants in National University Entrance Exam (Math track)
- Silver Medal – National Astronomy & Astrophysics Olympiad
- Bronze Medal – National Astronomy & Astrophysics Olympiad
- Financial scholarship for academic excellence (INEF)
- Member – Young Scholars' Club

Additional Activities

Organizing Committee Member – National Observatory Conference (2022)

Olympiad Instructor – Astronomy & Astrophysics, Salam High School (2017)